



DA 2310: *Data Engineering at Scale*

Course Project Presentation

Customer Satisfaction for Transport Service Application – UBER

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Problem Statement

- Objectives of the project
 - Identify the key factors influencing overall customer satisfaction and ride experience.
- Data source: <https://www.kaggle.com/datasets/yashdevladdha/uber-ride-analytics-dashboard>
- Description of the data (features, 40 MB, Tabular)
 - Features:
 - ['Date',' Time',' Booking ID',' Booking Status',' Customer ID',' Vehicle Type',' Pickup Location',' Drop Location',' Avg VTAT',' Avg CTAT',' Cancelled Rides by Customer',' Reason for cancelling by Customer',' Cancelled Rides by Driver',' Driver Cancellation Reason',' Incomplete Rides',' Incomplete Rides Reason',' Booking Value',' Ride Distance',' Driver Ratings',' Customer Rating',' Payment Method']
- The following tools are used for the project
 - **PySpark**
 - Pandas, Matplotlib/Seaborn, Scikit-learn
 - RandomForestClassifier, GBTClassifier, LogisticRegression
 - The project was originally developed and executed on **Google Colab** owing to higher memory requirement and later the necessary **.ipynb** files were uploaded the github link https://github.com/umeshbp-iisc/de_Uber_analysis/tree/main for the repository management.

SYSTEM ARCHITECTURE

DATA INGESTION LAYER

CSV DATA (NCR RIDES)

DATA PROCESSING LAYER (APACHE SPARK)

SparkSession Configuration
4GB Driver Memory, 4GB
Executor, 2 Executor
Cores, 200 Shuffle
Partitions

ETL Pipeline
Data Loading & Schema
Validation
Timestamp Merging (Date
+ Time)
Null Value Handling
Type Casting (String →
Double)
Type Casting (String →
Double)

DIMENSIONAL DATA MODEL

dim_customer
Customer ID
Ratings

dim_vehicle
Vehicle Type

dim_location
Pickup Loc
Drop Loc

facts_rides (Fact Table)
Customer ID , Vehicle Type ,
Pickup Location , Avg CTAT
RideCompleted (Target)

EDA

Pandas DataFrame
Conversion

Matplotlib/Seaborn Viz

Temporal Patterns

Booking Status
Distribution

Cancellation Analysis

ML Pipeline

Feature Engineering

String Indexing

Vector Assembly

Train/Test Split (70/30)

Model Ensemble
RandomForest
GBClassifier
LogisticRegression

MODEL EVALUATION

Accuracy: 95.57%

RMSE: 0.21

Cross-validation Ready

Sc

Data Collection & Feature Engineering

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 150000 entries, 0 to 149999
Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Date                                  150000 non-null object
1   Time                                  150000 non-null object
2   Booking ID                           150000 non-null object
3   Booking Status                       150000 non-null object
4   Customer ID                          150000 non-null object
5   Vehicle Type                         150000 non-null object
6   Pickup Location                     150000 non-null object
7   Drop Location                       150000 non-null object
8   Avg VTAT                            139500 non-null float64
9   Avg CTAT                            102000 non-null float64
10  Cancelled Rides by Customer          10500 non-null float64
11  Reason for cancelling by Customer    10500 non-null object
12  Cancelled Rides by Driver            27000 non-null float64
13  Driver Cancellation Reason           27000 non-null object
14  Incomplete Rides                    9000 non-null float64
15  Incomplete Rides Reason              9000 non-null object
16  Booking Value                       102000 non-null float64
17  Ride Distance                       102000 non-null float64
18  Driver Ratings                      93000 non-null float64
19  Customer Rating                     93000 non-null float64
20  Payment Method                      102000 non-null object
dtypes: float64(9), object(12)
memory usage: 24.0+ MB
```

```
input_df = input_df.withColumn("Timestamp",
                               to_timestamp(concat_ws(" ", col("Date"), col("Time")), "yyyy-MM-dd HH:mm:ss"))
input_df = input_df.withColumn("HourOfDay", hour(col("Timestamp")))
input_df = input_df.withColumn("DayOfWeek", date_format(col("Timestamp"), "EEEE"))
```

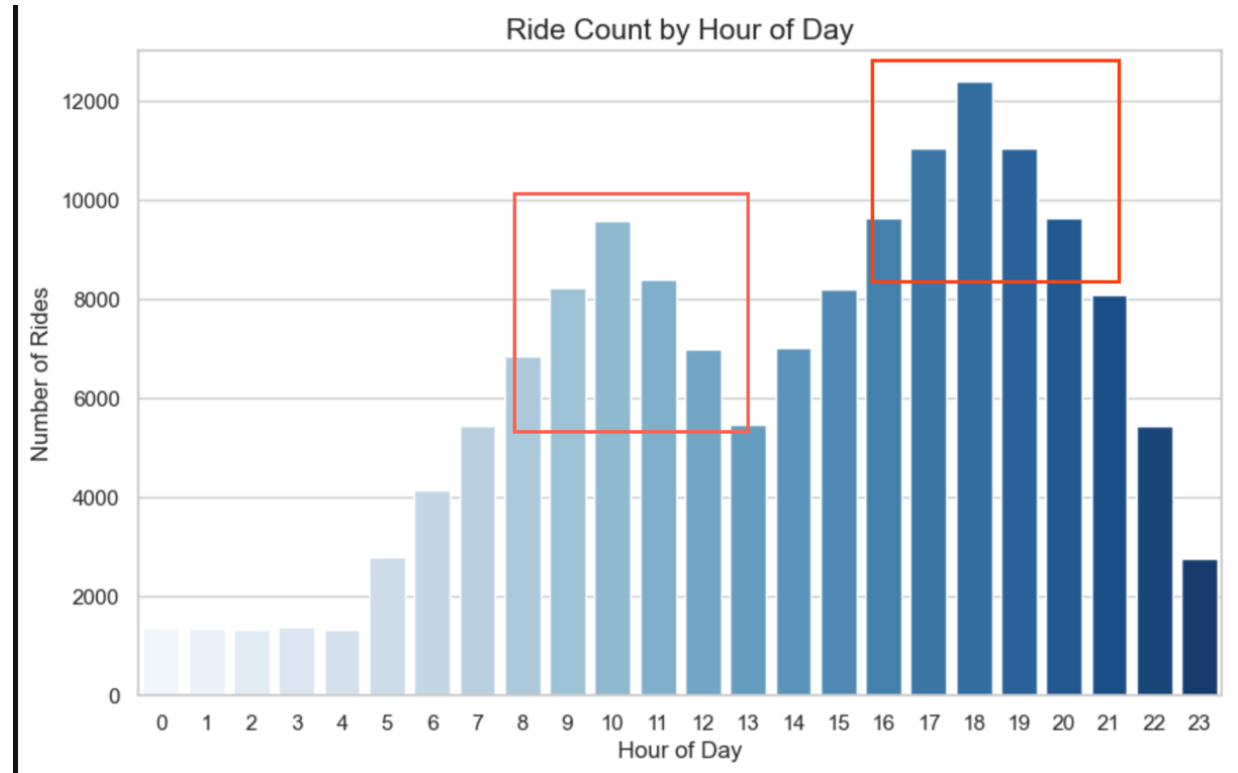
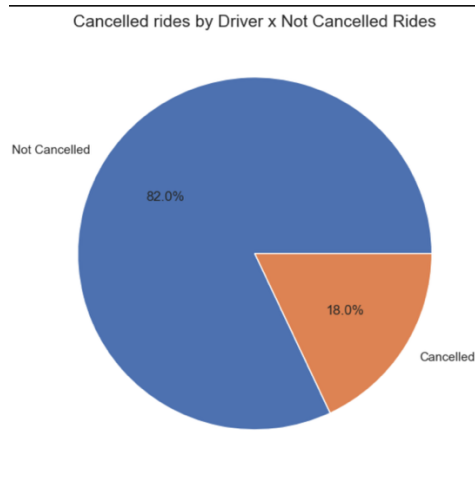
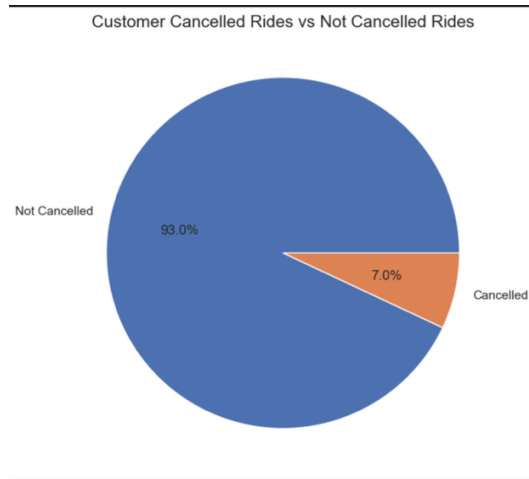
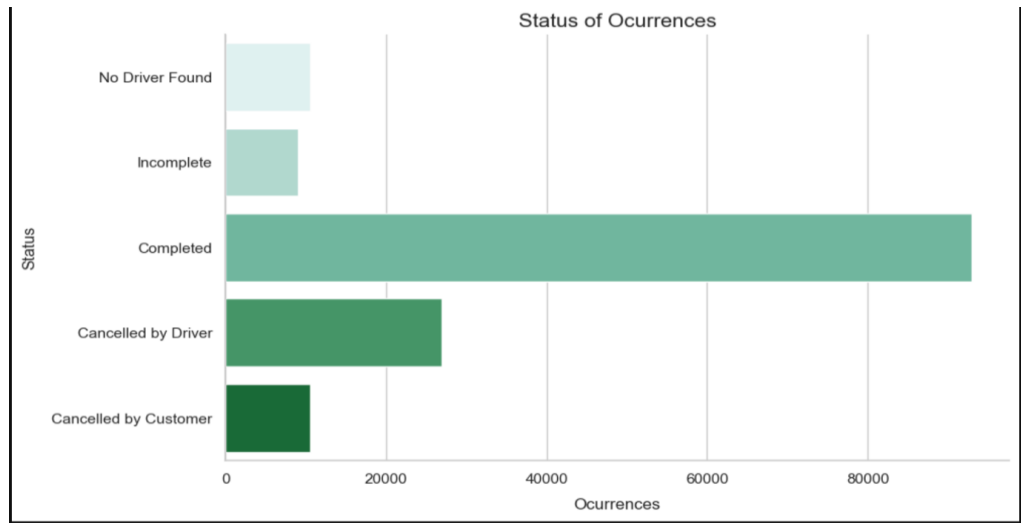
```
indexers = [
    StringIndexer(inputCol="Vehicle Type", outputCol="VehicleType_Index"),
    StringIndexer(inputCol="Pickup Location", outputCol="PickupLocation_Index"),
    StringIndexer(inputCol="Drop Location", outputCol="DropLocation_Index")
]
```

```
# Replace 'null' strings with actual nulls, then cast to double
#VTAT : Vehicle turn around time
#CTAT : Customer turn around time
numeric_columns = ['Avg VTAT', 'Avg CTAT', 'Booking Value', 'Ride Distance', 'Driver Ratings', 'Customer Rating']

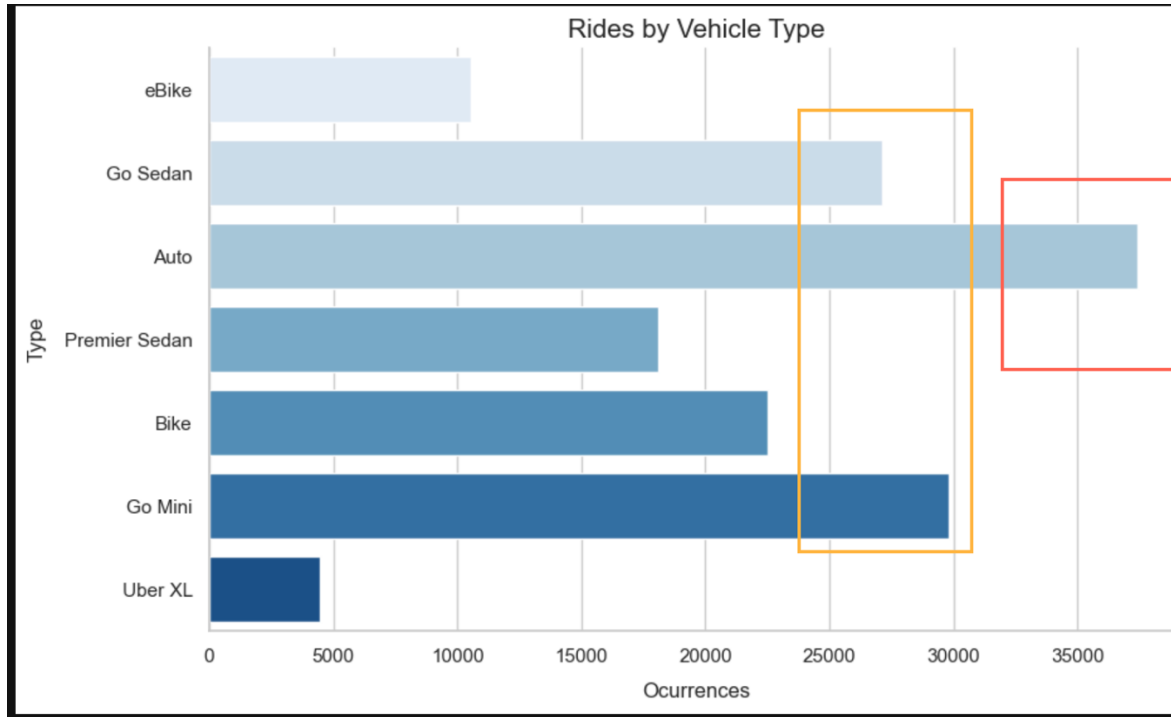
for column in numeric_columns:
    if column in input_df.columns:
        # Replace 'null' string with None, then cast to double
        input_df = input_df.withColumn(column,
                                         when(col(column) == "null", None)
                                         .otherwise(col(column))
                                         .cast("double"))

#Data cleaning: Fill null with default values
input_df = input_df.fillna({
    'Avg VTAT': 0.0,
    'Avg CTAT': 0.0,
    'Booking Value': 0.0,
    'Ride Distance': 0.0,
    'Driver Ratings': 0.0,
    'Customer Rating': 0.0
})
input_df.show()
```

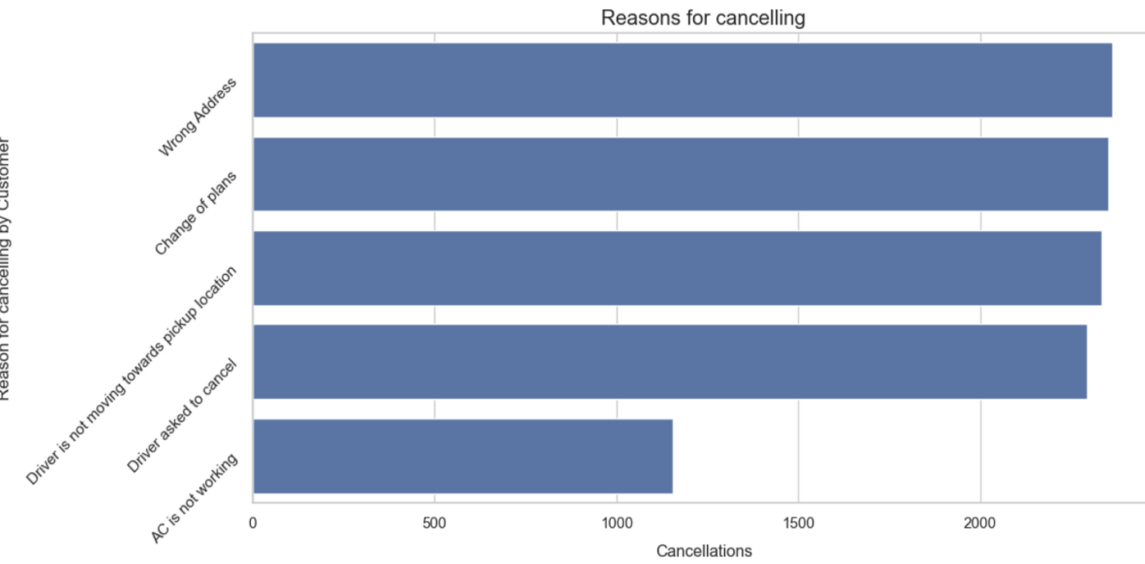
Exploratory Data Analysis (EDA)



Exploratory Data Analysis (EDA)



Reason for cancelling by Customer



Data Leakage Identification & Resolution , Final Feature Set

- **Critical Discovery:** Initial models showed perfect accuracy (1.0000) due to data leakage.
- **Leakage Features Identified:**
 - **Payment Method:** "Cancelled or incomplete ride" → 100% indicates non-completion
 - **Driver Ratings = 0:** Only available post-ride; 0 for all non-completed rides
 - **Customer Rating = 0:** Same issue as driver ratings
 - **Booking Value = 0:** Revenue only recorded for completed rides
- **Resolution:** Removed all post-event features, retaining only pre-booking information.
- **Numeric Features (3):**
 - Avg VTAT (Vehicle Turn Around Time)
 - Avg CTAT (Customer Turn Around Time)
 - Ride Distance
- **Categorical Features (3):**
 - Vehicle Type (indexed)
 - Pickup Location (indexed)
 - Drop Location (indexed)
- **Total Features:** 6 predictive attributes

Model Selection Rationale

- RandomForestClassifier
- **Advantages for Large-Scale Data:**
 - **Parallelizable:** Each tree trains independently across cluster nodes
 - **Handles Non-linearity:** Captures complex feature interactions
 - **Robust to Outliers:** Ensemble averaging reduces noise impact
 - **Feature Importance:** Built-in ranking for interpretability
 - **No Feature Scaling Required:** Works with raw categorical indexes
- **Scalability:** $O(n \log n \times \text{features} \times \text{trees})$ distributes well across Spark executors
- **Performance:** 95.57% Accuracy, 0.2104 RMSE

```
RandomForestClassifier(  
    numTrees=50,  
    maxDepth=10,  
    maxBins=200,  
    seed=42  
)
```


Model Selection Rationale

- **GBClassifier (Gradient Boosted Trees)**

- **Advantages for Large-Scale Data:**

- **Sequential Optimization:** Each tree corrects previous errors
 - **Handles Imbalance:** Naturally focuses on difficult examples
 - **High Accuracy:** Often achieves best performance in competitions
 - **Regularization:** Built-in mechanisms prevent overfitting

```
GBClassifier(  
    maxIter=50,  
    maxBins=200  
)
```

- **Scalability Consideration:** Less parallelizable than RandomForest (sequential nature), but Spark MLlib optimizes tree-level operations

- **Performance:** 95.49% Accuracy, 0.2124 RMSE

- **LogisticRegression**

- **Advantages for Large-Scale Data:**

- **Extremely Fast Training:** Linear complexity $O(n \times \text{features} \times \text{iterations})$
 - **Low Memory Footprint:** Minimal model size for deployment
 - **Interpretable Coefficients:** Clear feature contribution understanding
 - **Online Learning Capable:** Can update with streaming data
 - **Baseline Model:** Validates whether complex models add value

```
LogisticRegression(  
    maxIter=50  
)
```

- **Scalability:** Excellent - distributes gradient computation across partitions

- **Performance:** 94.70% Accuracy, 0.2303 RMSE

Pipeline Architecture

- **Benefits:**
 - **Reproducibility:** Entire workflow captured in single object
 - **Prevent Leakage:** Transformations fit only on training data
 - **Easy Deployment:** Save/load complete pipeline
 - **Hyperparameter Tuning Ready:** Compatible with CrossValidator
- **Training Process**
 - **Data Split:**
 - Training: 70% (105,000 rides)
 - Testing: 30% (45,000 rides)
 - Seed: 42 (reproducibility)
 - **Training Time:**
 - RandomForest: ~24 seconds
 - GBTCClassifier: ~32 seconds
 - LogisticRegression: <1 second
 - **Cluster Utilization:** Distributed across 200 partitions with 4GB memory allocation
- **Streamlit Link:** <https://teamdatacraftapp.streamlit.app/>

```
Pipeline(stages=[  
    StringIndexer(Vehicle Type),  
    StringIndexer(Pickup Location),  
    StringIndexer(Drop Location),  
    VectorAssembler(all features),  
    Classifier (RF/GBT/LR)  
])
```

Scalability – Performance with Big Data

Model	Runtime across multiple sizes of data		
	150k rows	1.6M rows (11x)	4.9M rows (33x)
RandomForest	9.97 seconds	43.65 seconds	133.72
GBT Classifier	23.62 seconds	131.78 seconds	508.11 seconds
Logistic Regression	4.83 seconds	20.01 seconds	78.98 seconds
Total Training time	38.41 seconds	195.44 seconds	720.81 seconds
Scalability Performance	1x	5x	18.7x

Model Interpretation

RandomForest - Best Overall Performance:

- 95.57% accuracy means ~44,000 correct predictions on 45,000 test rides
- RMSE 0.2104 indicates average prediction error of 21%
- Strong balance between accuracy and generalization

Why RandomForest Won:

- 1.Handles feature interactions (e.g., Vehicle Type + Pickup Location)
- 2.Robust to the ~176 unique location categories
- 3.Naturally handles class imbalance (62% completed vs 38% not completed)

Model	Accuracy	RMSE	Training Time	Model Size
RandomForest	95.57%	0.2104	24s	Large
GBTClassifier	95.49%	0.2124	32s	Medium
LogisticRegression	94.70%	0.2303	<1s	Small

Business Insights and Recommendation

- Deploy more drivers during peak hours (7-9 AM, 5-8 PM)
- Focus on high cancellation pickup locations
- Balance vehicle type availability by demand patterns
- Reduce wait times (primary cancellation reason)
- Implement dynamic pricing alerts
- Proactive ETA updates to reduce cancellations
- Use 95.57% accurate model to identify "at-risk" bookings
- Driver incentives for high-risk rides
- Target premium hours (late night) with better driver coverage
- Optimize vehicle mix based on demand patterns
- Reduce cancellation related revenue loss

Demo (followed by Q&A)